Leong Wai Men

Game & Engine Programmer

Seeking a Game Client / Engine Programming Internship

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Game & engine programmer studying Computer Science (Real-Time Interactive Simulation) at DigiPen Institute of Technology Singapore, graduating September 2028. Tech Lead and Engine Champion on Umapyoi Engine, a custom C++ engine our team of 7 used to ship Senki, a 2D hack-and-slash.

Skills

Languages: C++, C#, C, js, Lua

Engine & graphics: OpenGL, ImGui, Sol2, Unity

Online & data: PlayFab, Photon (PUN2), MySQL

Tools: Git / GitHub, Trello, Docker, Nginx

Team communication & conflict management

Projects

AnytimeReno / Web-Based 3D Interior Planner (Unity WebGL)

TECH LEAD & ENGINE CHAMPION · DIGIPEN · TEAM OF 7 · [ONGOING]

- Leading technical direction of a browser-based Unity/WebGL interior planner that lets homeowners lay out rooms and furnish them in 3D, backed by a self-hosted Node.js/MariaDB server.
- Editor architecture: Built a data-driven 2D floor plan editor where edits change only the plan data (nodes, edges, rooms), and a dirty-flag rebuild pass redraws walls from a pool of LineRenderers instead of creating objects per edit.
- Asset streaming: Implemented runtime glTF (.glb) loading into the WebGL client via glTFast, with UUID-keyed models and thumbnails served from the server.
- Backend & storage: Built a Node.js API and MariaDB database in Docker Compose behind nginx/HTTPS, with SQL migrations, automated backups, and per-user project and model storage with owner-scoped uploads.
- Auth & security: Implemented account registration and login with Argon2 password hashing and hashed session tokens, hardened with nginx per-route rate limiting and a CAPTCHA overlay on the WebGL canvas.
- CI/CD: Set up a GitHub Actions pipeline on a self-hosted runner that builds the Unity WebGL app and ships the API as a container image to the production VM.
- Website: Link

Umapyoi Engine & Senki / Custom C++ Game Engine

TECH LEAD & ENGINE CHAMPION · DIGIPEN · TEAM OF 7 · [6MONTHS]

- Led technical direction of a custom C++/OpenGL engine used by a 7-person team to ship *Senki*, a 2D top-down hack-and-slash.
- **Scripting:** Built a Lua scripting system (Sol2) that lets designers write gameplay and enemy behaviour scripts without touching engine code; scripted the Fire enemy's behaviour in Lua.
- **Editor & tools:** Built an in-engine tilemap system and editor (ImGui) that designers used to build the game's levels, with editor undo/redo.
- **Prefabs:** Built a prefab system supporting nested prefabs, resolving conflicts where scene data overwrote prefab data and nested prefabs broke each other's dependencies.
- **Architecture:** Designed the ECS core (entity manager, components, systems) that all engine and gameplay code runs on.
- **Gameplay systems:** Implemented collision/physics and controller input; integrated a PlayFab online leaderboard.
- Game: Link
- Engine: Link

Droom

PATH FINDING & FRONTEND · DIGIPEN · TEAM OF 7 · [3MONTHS]

- A Doom-style first-person shooter rendered in 2.5D with raycasting. The 3D view is built entirely from a 2D grid map, one screen column at a time.
- Wave system controlling enemy spawns, wave progression and difficulty scaling
- Grid-based enemy pathfinding with A* so enemies navigate the map to reach the player
- Game UI: HUD for health and ammo, wave counter, and menu screens
- Game: Link

Death Dungeon / 2D Multiplayer Action Mobile Game

OCTOBER 2021 - JANUARY 2022, SINGAPORE

- Fast-paced arena fighting game for Android, built solo in Unity.
- Online multiplayer with a self-developed lag compensation feature.
- Player login, including Google / Facebook login, with player data stored in a locally hosted MySQL database.

Escape From Dungeon / 2D Local Multiplayer Platformer

OCTOBER 2021 - JANUARY 2022, SINGAPORE

- Year 2 studio team project built in Unity for Windows and WebGL.
- Built local 2-player multiplayer on a single computer.
- Built enemy AI using a state machine.
- Built the audio manager and game UI (main menu, pause screen).

Experience

Metascape City / Game Developer
MAY 2022 - AUGUST 2022, SINGAPORE

- Worked on the early stage of a multiplayer WebGL app where users explore a virtual recreation of Singapore and interact with each other.
- Added 3D buildings to the environment and optimised 3D model rendering performance using BVH-based techniques.
- Implemented minor multiplayer interactions between users.

VizioFly / Software Developer (Internship)
MARCH 2021 - AUGUST 2021, SINGAPORE

- Built interactive Unity applications for clients, including a Virtual Expo where users walk around an open area and interact with booths.
- Built an Online Presentation feature for hosting presentations on a WebGL website.
- Implemented Photon (PUN2) multiplayer, UI, NavMesh navigation and Facebook integration in Unity.
- Built server-side features in PHP, HTML and JavaScript, using MySQL and PlayFab with Facebook login for user data and authentication.

Ritz-Carlton Hotel / Banquet Server (Part-time)
DURING SCHOOL HOLIDAYS, SINGAPORE

- Served guests as part of a banquet team at dinner events, and led small groups to complete assigned tasks.

Education

DigiPen Institute of Technology Singapore & Singapore Institute of Technology / BSc Computer Science in Real-Time Interactive Simulation
AUGUST 2024 - SEPTEMBER 2028 (EXPECTED), SINGAPORE

- Relevant coursework: Data Structures & Algorithms, Computer Networks, Computer Graphics, Linear Algebra, Discrete Mathematics

Nanyang Polytechnic / Diploma in Game Development & Technology
MARCH 2019 - MARCH 2022, SINGAPORE

- Distinction in Studio Project (Game Project)
- Coursework: OpenGL / C++ graphics, physics, server & database, multiplayer and console game development, level design